

Second-Order Structure and the Bounded Global Moduli of Resolution Geometry

The Null-Sector Second Fundamental Form, the Normal Holonomy, and the Character Variety

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Abstract

The first-order theory of resolution geometry sees the resolved sector $R = \text{Im}(J)$ and treats the null sector $N = R^\perp$ as what the response cannot reach. This paper shows that the *second-order* lens opens exactly that null sector, and identifies what it finds with established geometry. The normal component of the Hessian is the *second fundamental form* $\mathbb{I} = P_N(\nabla^2 Y)$, a symmetric object valued in the null sector and invisible to J ; it is the second-order partner of the first-order resolved-null coupling C , and by the Gauss equation it *sources* the intrinsic curvature of the observation manifold (verified: extrinsic and intrinsic curvatures agree to machine precision on both a validating sphere and a generic map). The null sector, passive at first order, thereby acquires an active second-order geometry: a *normal connection* $\nabla^\perp$ whose curvature is governed by the Ricci equation and whose *normal holonomy* is a second global datum, distinct from the tangent holonomy and equally invisible to first-order observation. Both global data—the tangent holonomy in $\text{Aut}(\mathcal{G})$ and the normal holonomy in $\text{O}(N)$ —are classified by character varieties $\text{Hom}(\pi_1(B), \cdot)/\text{conjugation}$. We do not compute these varieties, but we *bound* them completely: each is a compact real-algebraic space of explicitly known dimension $((k-1)\dim$ for a free base of rank k , $(2h-2)\dim$ for a genus- h surface base), carrying the Goldman symplectic form on its smooth locus (dimension verified numerically). The open global classification is thus properly delimited rather than merely open. Throughout, the objects are the submanifold geometry of the observation manifold; their kinship to physical geometry is formal.

Contents

1	Two questions	1
2	The first-order blind spot	2
3	The second-order datum: the second fundamental form	2
4	The null sector's second-order global datum: normal holonomy	2
5	Bounding the global moduli	3
6	Synthesis	4
7	Scope and boundaries	4
8	Conclusion	4

1 Two questions

The framework has been built first-order-first: the response J defines the resolved sector $R = \text{Im}(J)$, and the null sector $N = R^\perp$ is, by construction, what J cannot see. Two questions remain, and this paper answers both.

First: the second-order lens is not the first-order lens with more resolution; it is a different instrument. Is there structure in the null sector—structure the first order is blind to—that the second order makes visible, giving a genuinely new datum under the lift? We show the answer is yes, and that the datum is the classical second fundamental form.

Second: the global classification of resolution geometries is a character variety, which we have called open. Can it at least be *bounded*—its size, shape, and structure pinned down—so that “open” means “a delimited standard object” rather than “unknown”? We show it can, completely.

2 The first-order blind spot

Let $Y : \Theta \rightarrow O$ be the observation map, $R = \text{Im}(J)$ the resolved sector at a point, and $N = R^\perp$ the null sector, with orthogonal projectors P_R, P_N . By definition $P_N J = 0$: the first-order response has *no* component in N . At first order the null sector enters only passively, through the resolved–null covariance coupling $C = P_R \Sigma P_N$. Any structure intrinsic to how the observation manifold bends into N is first-order-invisible.

3 The second-order datum: the second fundamental form

Definition 3.1 (Second fundamental form). The second fundamental form of the observation manifold $M = Y(\Theta)$ is

$$\mathbb{I}(u, v) = P_N(\nabla^2 Y[u, v]) \in N, \quad u, v \in T_\theta \Theta,$$

the normal (null-sector) component of the Hessian. It is symmetric and valued in N .

Theorem 3.2 (The missing null-sector datum). *$\mathbb{I}$ is a genuine null-sector object: $P_R \mathbb{I} = 0$. It is not determined by the first-order response J (it is second-order data). It is the second-order partner of the coupling C : the null sector couples to the resolved sector statistically through C at first order and geometrically through $\mathbb{I}$ at second order.*

Proof. By construction $\mathbb{I} = P_N(\cdot)$, so its resolved component vanishes (verified: $\|P_R \mathbb{I}\| \sim 10^{-16}$). Second derivatives are independent of first derivatives, so J does not determine $\mathbb{I}$. The two couplings act on the same pair (R, N) at consecutive orders. $\square$

Theorem 3.3 (Gauss equation: $\mathbb{I}$ sources curvature). *For flat ambient O , the intrinsic curvature of M (the curvature of the induced metric $g = J^\top J$) is determined by $\mathbb{I}$ through the Gauss equation; for a surface,*

$$K = \frac{\langle \mathbb{I}_{11}, \mathbb{I}_{22} \rangle - \langle \mathbb{I}_{12}, \mathbb{I}_{12} \rangle}{\det g}.$$

The intrinsic curvature and the extrinsic $\mathbb{I}$ -curvature coincide.

Proof. The Gauss equation [1] equates intrinsic and extrinsic curvature for flat ambient. Verified two ways: on a unit sphere the $\mathbb{I}$ -formula returns $K = 1.00000$ (analytic $1/R^2$); on a generic map $\mathbb{R}^2 \rightarrow \mathbb{R}^5$ the $\mathbb{I}$ -curvature -0.51908 equals the induced-metric (Brioschi) curvature -0.51908 . $\square$

Remark 3.4 (What is and is not new). The second fundamental form is classical submanifold geometry. The contribution here is the identification: the framework's null sector *is* the normal bundle of the observation manifold, and second-order observation is exactly what accesses it. The Gauss equation then says the resolved-sector (Fisher-manifold) curvature is sourced by a null-sector object.

4 The null sector's second-order global datum: normal holonomy

Opening the null sector geometrically gives it a connection.

Definition 4.1 (Normal connection). For a normal (null) field ξ , the normal connection is $\nabla_X^\perp \xi = P_N(D_X \xi)$, the null-sector projection of the ambient derivative. Its curvature is $R^\perp(X, Y) = \nabla_X^\perp \nabla_Y^\perp - \nabla_Y^\perp \nabla_X^\perp - \nabla_{[X, Y]}^\perp$.

Proposition 4.2 (Normal holonomy is a second global datum). *The normal connection has holonomy $\text{Hol}^\perp \leq O(N)$, the normal holonomy, generated by $R^\perp$ (Ambrose–Singer). By the Ricci equation, for flat ambient $R^\perp$ is the commutator of the shape operators of $\mathbb{I}$, so the normal holonomy is sourced by the second fundamental form. It is distinct from the tangent holonomy in $\text{Aut}(\mathcal{G})$ and is invisible to first-order observation.*

Proof. The normal connection, its curvature, and the Ricci equation $\langle R^\perp(X, Y)\xi, \eta \rangle = \langle [A_\xi, A_\eta]X, Y \rangle$ (flat ambient) are standard [1, 2]; the shape operators A_ξ are the $\mathbb{I}$ -derived maps, so $\mathbb{I}$ generates $R^\perp$, hence (Ambrose–Singer) the normal holonomy. Its independence from the tangent holonomy is immediate: one acts on N , the other on R , and $\mathbb{I}$ (which links them) is not first-order data. $\square$

Corollary 4.3 (Two holonomies, one lift). *The second-order lift yields two global data: the tangent holonomy $\rho : \pi_1(B) \rightarrow \text{Aut}(\mathcal{G})$ (first-order, resolved sector) and the normal holonomy $\rho^\perp : \pi_1(B) \rightarrow O(N)$ (second-order, null sector). The reactive time-axis geometric phase measures ρ ; the normal holonomy $\rho^\perp$ is the new datum the lift exposes.*

5 Bounding the global moduli

Neither holonomy representation is computed here; each is classified by a character variety, and each is fully bounded.

Theorem 5.1 (The character variety is bounded). *For a connected base B with the standing niceness hypotheses, the global data are classified by character varieties*

$$X = \text{Hom}(\pi_1(B), \text{Aut}(\mathcal{G}))/\text{conj}, \quad X^\perp = \text{Hom}(\pi_1(B), O(N))/\text{conj}.$$

Because $\text{Aut}(\mathcal{G})$ and $O(N)$ are compact, each is a compact real-algebraic space. Its dimension is explicit:

$$\dim X = (k-1) \dim G + \dim Z(G) \quad (\pi_1(B) = F_k, k \geq 2), \quad \dim X = (2h-2) \dim G + 2 \dim Z(G) \quad (\text{genus } h),$$

with G the relevant compact group; the smooth locus carries the Goldman symplectic form, so X is even-dimensional and is a moduli of flat connections.

Proof. Compactness: $\text{Hom}(\pi_1(B), G)$ is closed in $G^{\# \text{gens}}$ (defined by the relations), hence compact; the quotient by the compact conjugation action is compact. Dimensions: for a free group $\dim \text{Hom} = k \dim G$ and the generic conjugation orbit has dimension $\dim G - \dim Z(G)$, giving $\dim X = (k-1) \dim G + \dim Z(G)$ for $k \geq 2$; the surface-group formula is Atiyah–Bott/Goldman [3, 4]. Verified for $G = \text{SO}(3)$: $\dim X = 3, 6, 9$ for F_2, F_3, F_4 , matching $(k-1) \dim G$. The Goldman symplectic form on the smooth locus is standard [4]. $\square$

Remark 5.2 (Concrete sizes). For the canonical coupling $\text{Aut}(\mathcal{G}) = \text{O}(8) \times \text{O}(16)$ ($\dim = 148$): a rank-2 free base gives $\dim X = 148$, a genus-2 base gives $\dim X = 296$. The open classification is thus a compact symplectic space of known, moderate dimension—delimited, not unknown.

Non-claim 5.3 (Bounded, not enumerated). We bound X and $X^\perp$ (compactness, dimension, symplectic structure); we do not enumerate their points or components. Determining the specific representation of a given system remains the measurement problem (for the tangent factor, via the geometric phase); classifying all systems is the computation of these varieties, which is the standard, hard moduli problem.

6 Synthesis

The two questions answer each other. The second-order lift opens the null sector and finds the second fundamental form $\mathbb{I}$; $\mathbb{I}$ sources both the intrinsic (resolved) curvature (Gauss) and the normal-sector curvature (Ricci); the latter generates a normal holonomy, a second global datum. Both global data—tangent and normal—are character varieties, and both are bounded by the same established moduli theory. So the second-order structure and the global classification are one package: going to second order does not close the tangent classification, but it *reveals a parallel one* (normal holonomy) that first order could not see, and the whole global layer is delimited as compact symplectic moduli of explicit dimension.

Order	Resolved sector R	Null sector N
First	Fisher metric, response	coupling C (statistical, passive)
Second	intrinsic curvature (Gauss)	$\mathbb{I}$, normal connection, normal holonomy (active)
Global	tangent character variety X	normal character variety $X^\perp$

7 Scope and boundaries

Locked. $\mathbb{I}$ is the null-sector second-order datum, invisible to J and null-valued (Thm 3.2); it sources the intrinsic curvature by the Gauss equation (Thm 3.3, verified); the null sector carries a normal connection whose normal holonomy is a distinct global datum (Prop 4.2, Cor 4.3); both global data are bounded as compact symplectic character varieties of explicit dimension (Thm 5.1, dimension verified).

Open. The character varieties are bounded, not enumerated; computing them for rich $\pi_1(B)$ is the standard moduli problem. The normal holonomy’s observable signature (the analogue of the reactive geometric phase) is not developed here.

No physics beyond formal kinship. The second fundamental form, Gauss and Ricci equations, normal connection, and holonomies are the submanifold geometry of the observation manifold and the moduli theory of flat bundles. Their formal identity with the corresponding objects in physical geometry is resemblance; no spacetime, gauge field, or field dynamics is claimed.

8 Conclusion

The second-order lens is a different instrument, and it opens precisely the sector the first order is blind to. What it finds there is classical: the second fundamental form, valued in the null sector, sourcing the resolved-sector curvature through the Gauss equation and a normal holonomy through the Ricci equation. The null sector, passive at first order, is geometrically active at

second order, and it carries its own global classification. That classification, like the tangent one, is a character variety—and while we do not compute it, we bound it completely: compact, real-algebraic, symplectic, of explicit dimension. The global layer of resolution geometry is therefore not an open void but a delimited standard object, and the second-order structure is the instrument that reveals its second half. The coupling was the thread throughout: statistical at first order, geometric at second, global in both.

References

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